

Math 1314 Lesson 21

1) Find the zeros of the polynomial function $f(x) = -2(x+1)(x-5)$.

a) Graph $f(x) = -2(x+1)(x-5)$. $\Rightarrow$

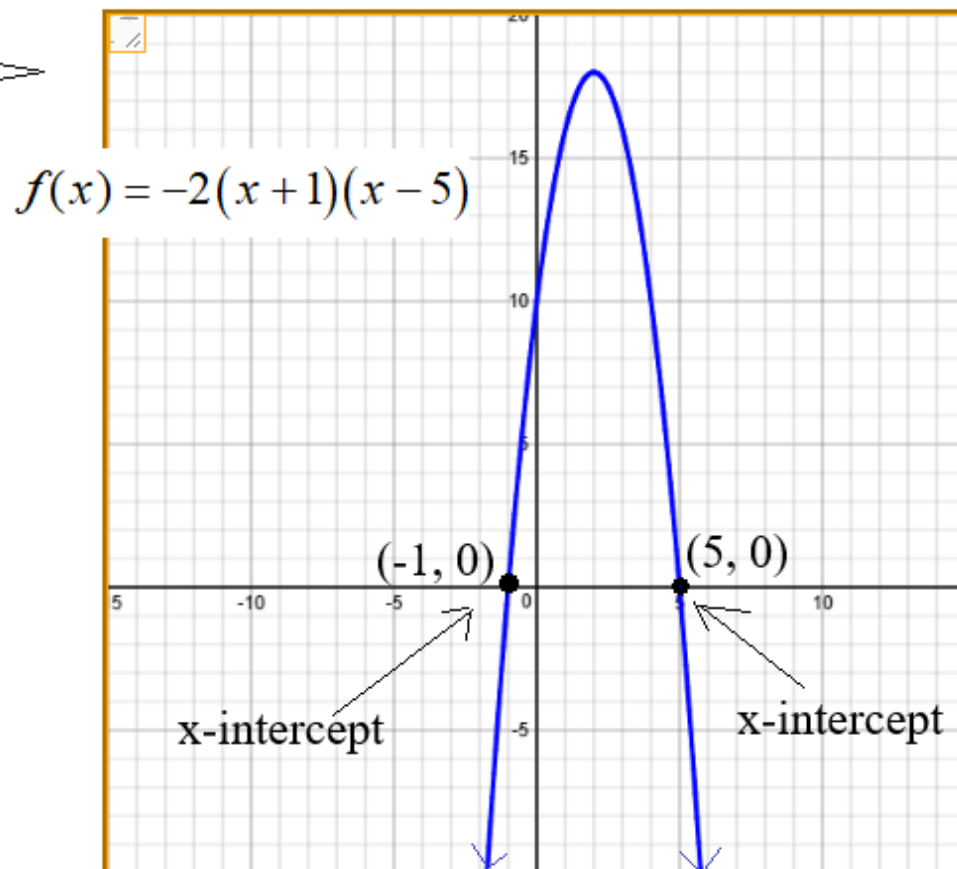
b) x-intercepts are _____

c) $f(-1) = \underline{\quad ? \quad}$

d) $f(5) = \underline{\quad ? \quad}$

e) Zeros of $f(x)$ are $\underline{\quad ? \quad}$

f) Domain of $f(x)$ is $\underline{\quad ? \quad}$



2) Find the zeros of the polynomial function $f(x) = -4(x + 3)^2(x - 6)^3$.

a) Graph $f(x) = -4(x + 3)^2(x - 6)^3$. $\Rightarrow$

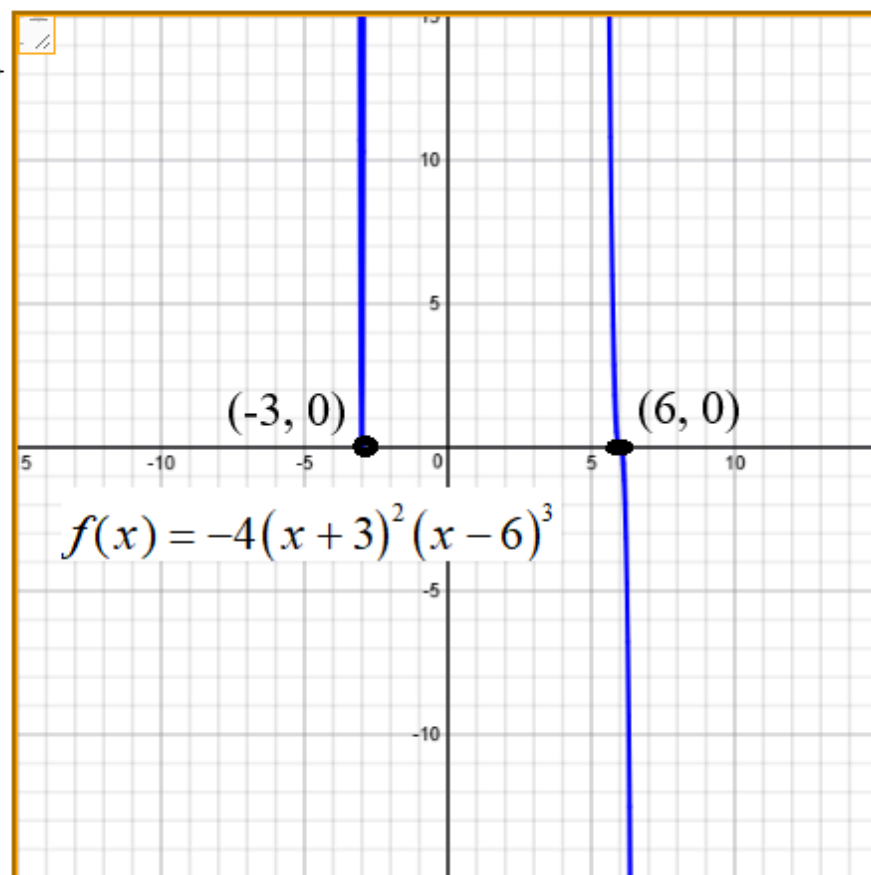
b) x-intercepts are _____

c) $f(-3) = \underline{\quad ? \quad}$

d) $f(6) = \underline{\quad ? \quad}$

e) Zeros of $f(x)$ are $\underline{\quad ? \quad}$

f) Domain of $f(x)$ is $\underline{\quad ? \quad}$



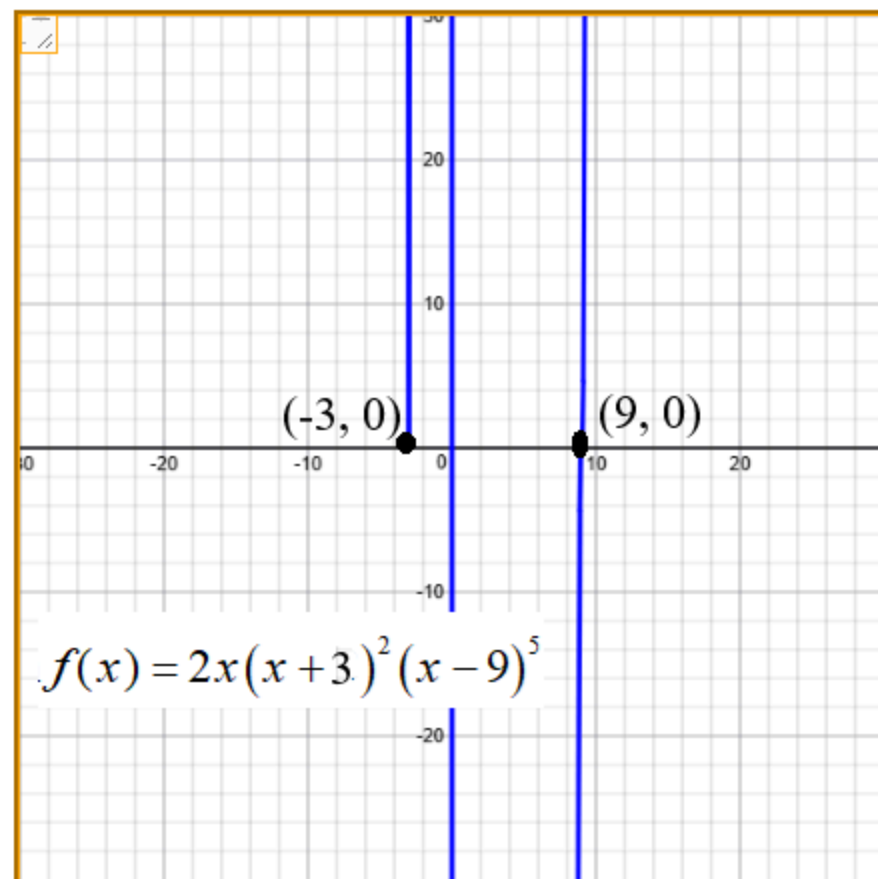
3) Find the zeros of the polynomial function $f(x) = 2x(x+3)^2(x-9)^5$.

a) $f(0) = \underline{\hspace{2cm}}?$

b) $f(-8) = \underline{\hspace{2cm}}?$

c) $f(9) = \underline{\hspace{2cm}}?$

d) Zeros of $f(x)$ are $\underline{\hspace{2cm}}?$

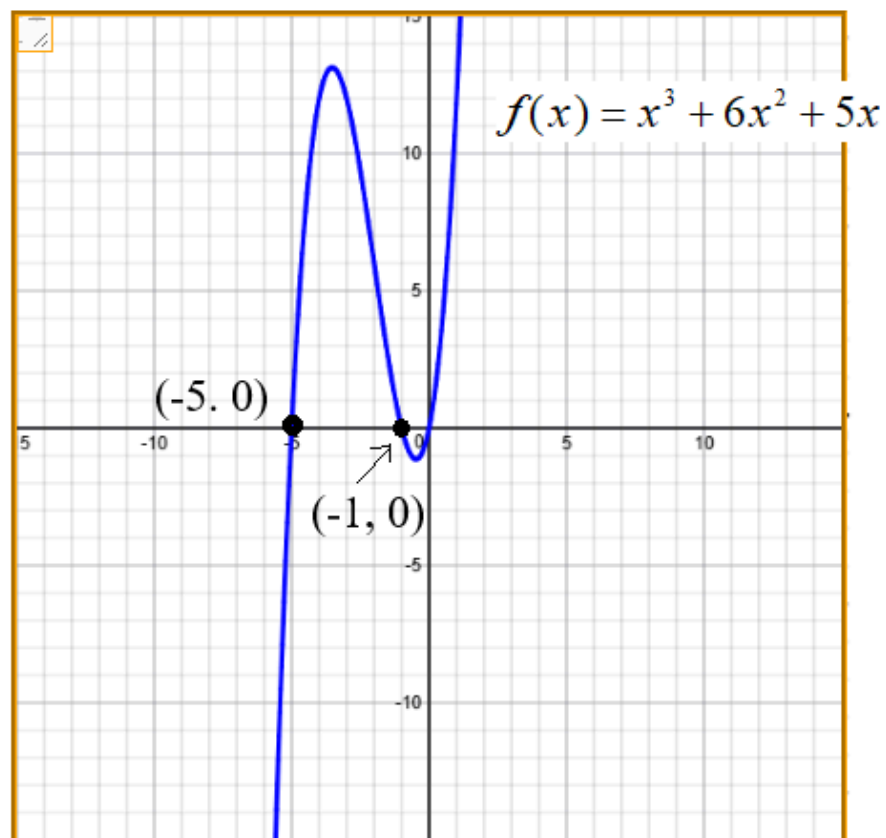


4) Find the zeros of the polynomial function $f(x) = x^3 + 6x^2 + 5x$.

a) Factor $x^3 + 6x^2 + 5x = x \cdot (\underline{\quad ? \quad}) \cdot (\underline{\quad ? \quad})$

b) Graph $f(x) = x^3 + 6x^2 + 5x$. 

c) Zeros of $f(x)$ are ?

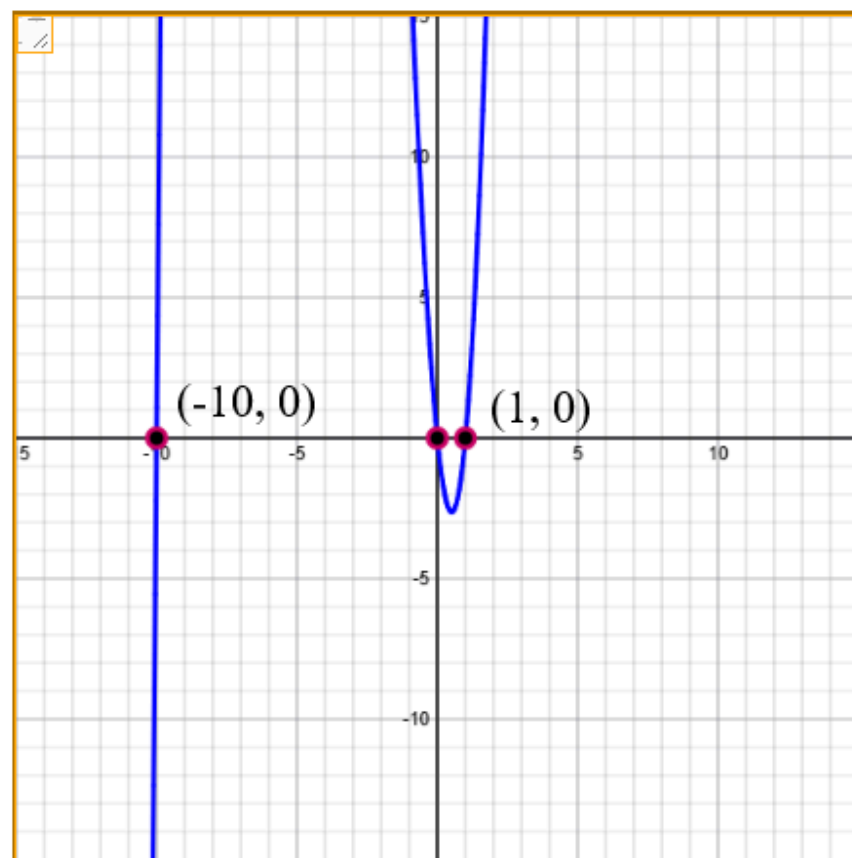


5) Find the zeros of the polynomial function $f(x) = x^3 + 9x^2 - 10x$.

a) Factor $x^3 + 9x^2 - 10x = x \cdot (\underline{\quad? \quad}) \cdot (\underline{\quad? \quad})$

b) Graph $f(x) = x^3 + 9x^2 - 10x$ 

c) Zeros of $f(x)$ are $\underline{\quad? \quad}$

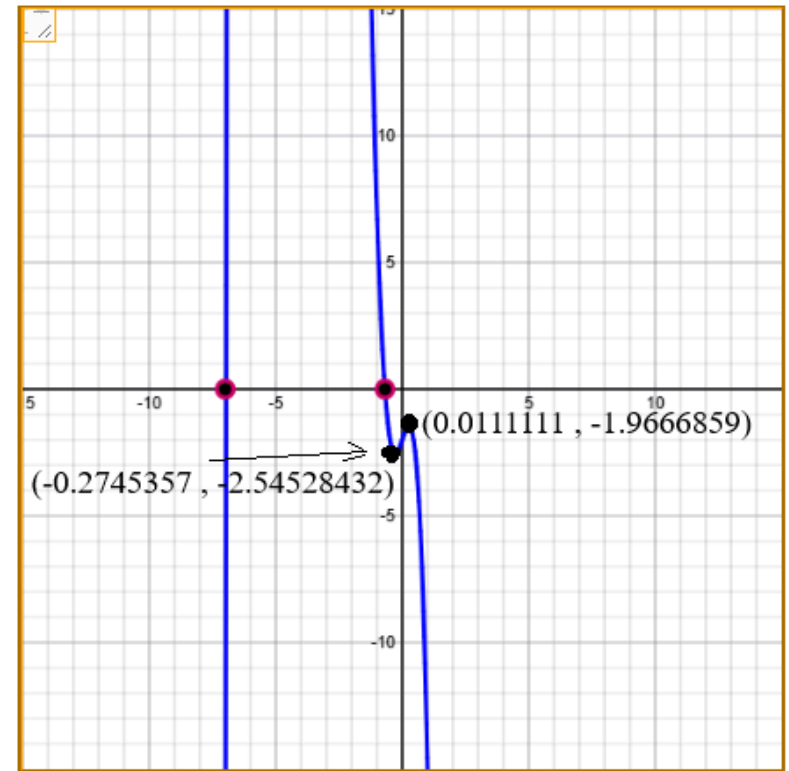


6) $f(x) = -2x^4 - 14x^3 + 3x - 2$

a) Graph $f(x) = -2x^4 - 14x^3 + 3x - 2$ 

b) Find the local minimum point between $x = -1$ and $x = 0$.

c) Find the local maximum point between $x = 0$ and $x = 1$.



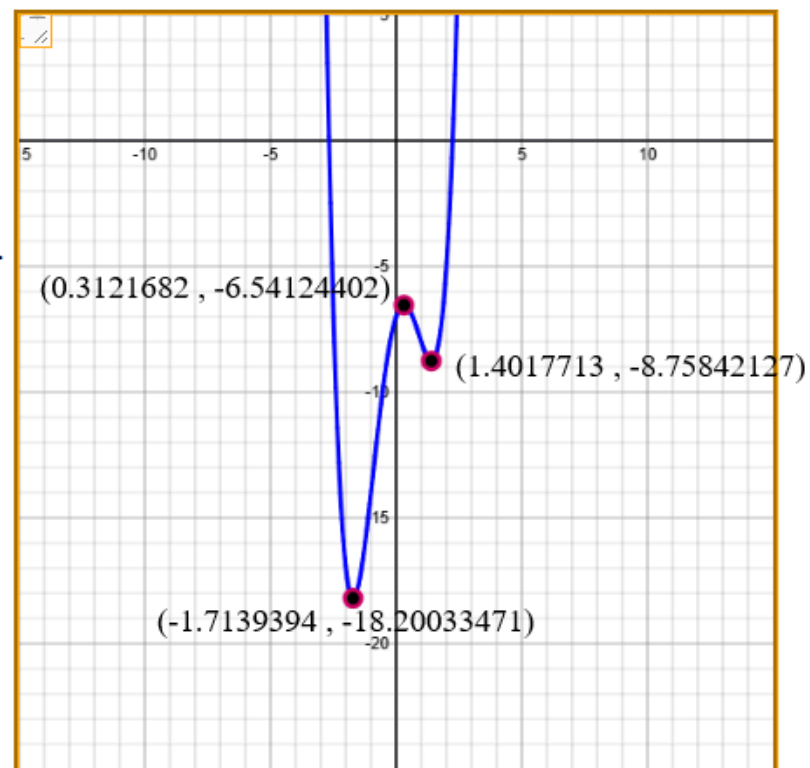
7) $f(x) = x^4 - 5x^2 + 3x - 7$

a) Graph $f(x) = x^4 - 5x^2 + 3x - 7$

b) Find the local minimum point between $x = -2$ and $x = -1$.

c) Find the local maximum point between $x = 0$ and $x = 1$.

d) Find the local minimum point between $x = 1$ and $x = 2$.



8) The revenue of a local TV station can be modeled by the polynomial function

$$R(t) = -0.04t^4 + 1.2t^3 - 17t^2 + 117.8t - 200,$$

where R represents the revenue in millions of dollars and t represents the year, with $t = 1$ corresponding to 2010.

Find the revenue for year 2015 (note: $t = 6$)

Find the revenue for year 2021 (note: $t = 12$)

